

Benjamin Ponce-Martinez

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EDUCATION

Iowa State University Ames, IA

May 2027

Bachelor of Science, Electrical Engineering GPA: 2.7

WORK EXPERIENCE

IINSPIRE LSAMP, Iowa State University Department of Mechanical Engineering

Aug 2022 – Aug 2023

Undergraduate Research Assistant

- Designed and fabricated acoustic devices integrating ultrasonic fields into additive manufacturing platforms for advanced 3D printing applications
- Developed and optimized an acoustic flow cell capable of extruding internally structured composite materials with improved energy efficiency
- Modeled and fabricated millimeter-scale fluid channels using 3D CAD design and stereolithography printing techniques for fluid mechanics research

IINSPIRE LSAMP VSEER, Iowa State University Department of Mechanical Engineering

May 2022 – Aug 2022

Undergraduate Research Assistant

- Designed and operated glass-based capillary micromodels to study oil displacement and porous media flow behavior under microscope imaging
- Improved micromodel fabrication processes to enhance durability and experimental reliability
- Prepared custom fluid solutions and performed controlled flow visualization experiments to optimize imaging clarity and data accuracy

PROJECTS

Temperature Indicator Circuit – EE230 Design Lab

Aug 2024 – Dec 2024

- Designed and built an analog temperature indicator circuit using an LM35 sensor and op-amp comparators to control multi-color LED outputs
- Implemented hysteresis to ensure stable LED transitions and accurate response to rising and falling temperatures
- Calibrated and validated circuit thresholds at 30°C, 40°C, 80°C, and 100°C using a variable DC voltage source

Active Audio Crossover Network – EE230 Design Lab

Aug 2024 – Dec 2024

- Designed and implemented a three-band active crossover circuit to separate audio input into low (<300 Hz), mid (300–5000 Hz), and high (>5000 Hz) frequency outputs
- Utilized op-amp-based second-order and first-order filter stages to achieve target roll-offs (40 dB/decade and 20 dB/decade) with adjustable low-frequency gain (1–5×)
- Characterized and validated circuit frequency response using signal generator and oscilloscope measurements to confirm accurate filter performance

Digital Logic Design – 4-Bit Prime Detector (CPRE 281)

Aug 2023 – Dec 2023

- Designed and debugged a 4-bit prime number detector using SOP, K-map simplification, and Verilog implementation
- Converted a NAND-only circuit into an equivalent AND/OR/NOT design and identified logical discrepancies
- Developed corrective logic to patch an existing circuit, optimizing functionality while minimizing added components

SKILLS

- Technical Skills: circuit designing, microsoft office, solidworks, python, java, C, verilog
- Soft Skills: leadership, time management, team player, attention to detail